1956 RSV discovered

1998 Palivizumab approved

2022 Nirsevimab trial

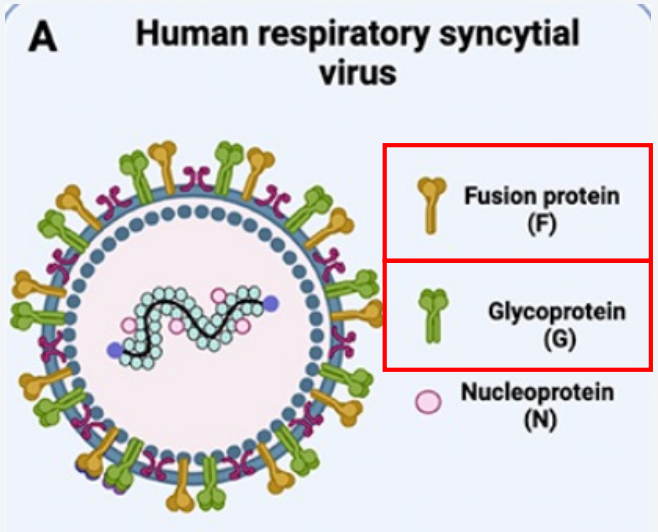
2023 Maternal RSV vaccine

RSV Prevention in Infants

Rebecca Mammarappallil

Respiratory Syncytial Virus

(RSV)



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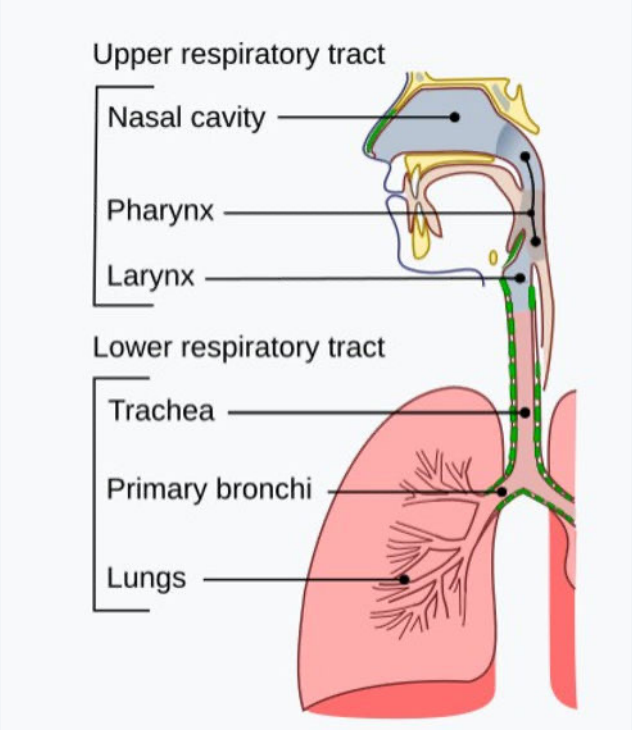
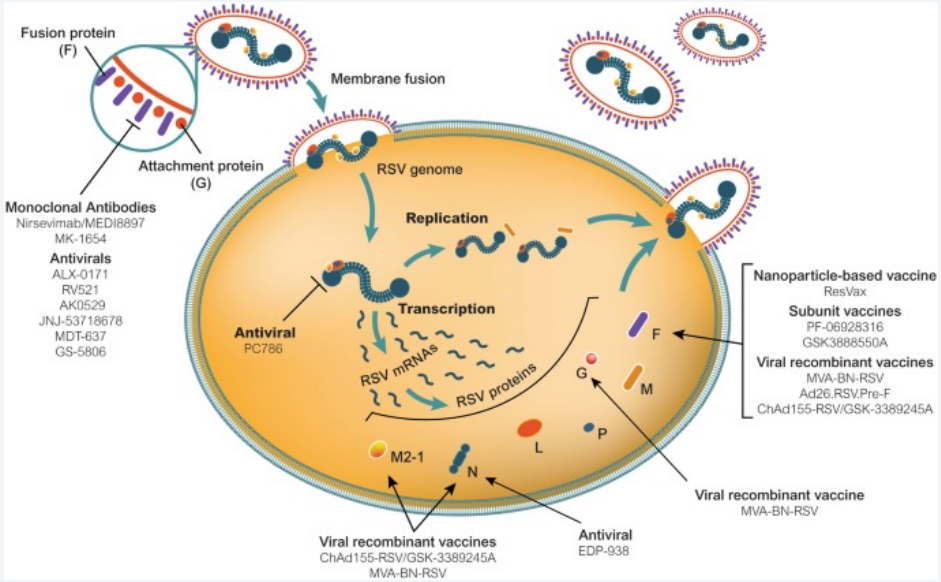


https://www.careers360.com/biology/respiratory-system-overview-topic-pge

Respiratory Syncytial Virus

(RSV)



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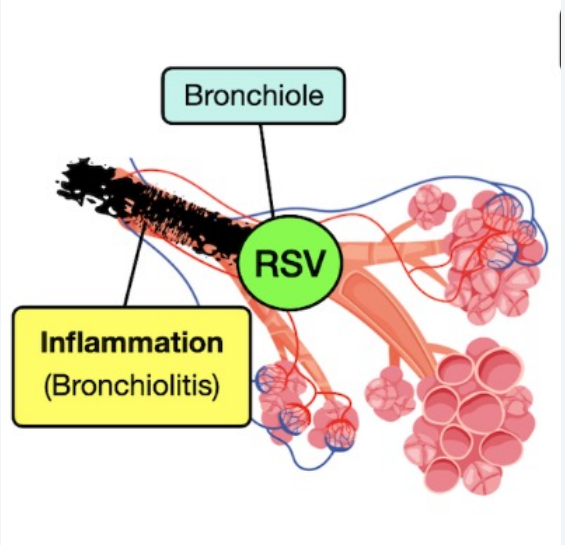
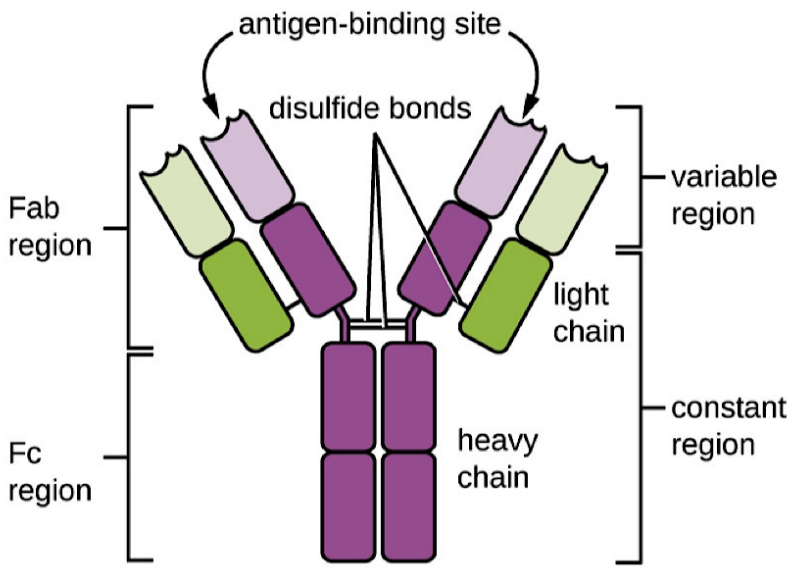


https://my.clevelandclinic.org/health/diseases/8272-bronchiolitis

Monoclonal Antibodies & IgG in RSV Prevention

IgG: The Key Antibody Class



https://me-pedia.org/wiki/Immunoglobulin_G

RSV-Targeted Therapeutics

RSV-IGIV (1990s)

Polyclonal IV immunoglobulin. Proved concept that passive antibody prevents severe RSV, but required monthly IV infusions of 15 mL/kg.

Palivizumab (1998)

Humanized IgG1 mAb binding RSV F-protein. Delivered IM monthly, a major advance, but limited to high-risk infants.

Nirsevimab (2022)

Recombinant IgG1κ with Fc YTE mutations extending half-life to ~69 days. Single dose covers an entire RSV season.

The RSV Prevention Timeline

Era 1: Passive Prophylaxis

Palivizumab (Synagis)

1998

First mAb proven effective in an infectious disease. Monthly IM injections for high-risk premature infants and those with BPD.

Era 2: Next-Gen mAb

Nirsevimab (Beyfortus)

2022

Extended half-life IgG1 with single-dose, season-long protection. Expanded to all infants, not just high-risk.

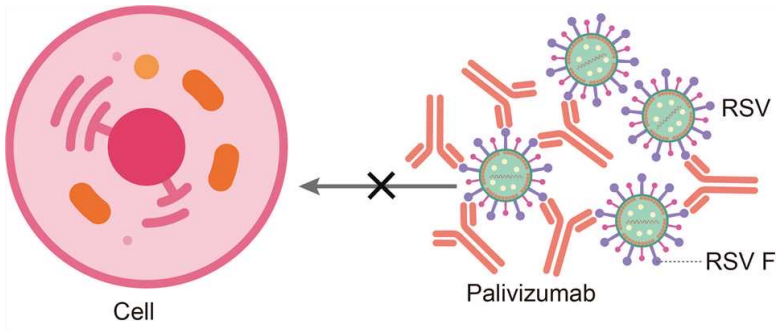
Era 3: Maternal Vaccine

RSVpreF (Abrysvo)

2023

Bivalent prefusion F vaccine given during pregnancy. Transplacental IgG transfer protects newborns from birth.

IMpact-RSV: Palivizumab



https://www.creativebiolabs.net/palivizumab-overview.htm

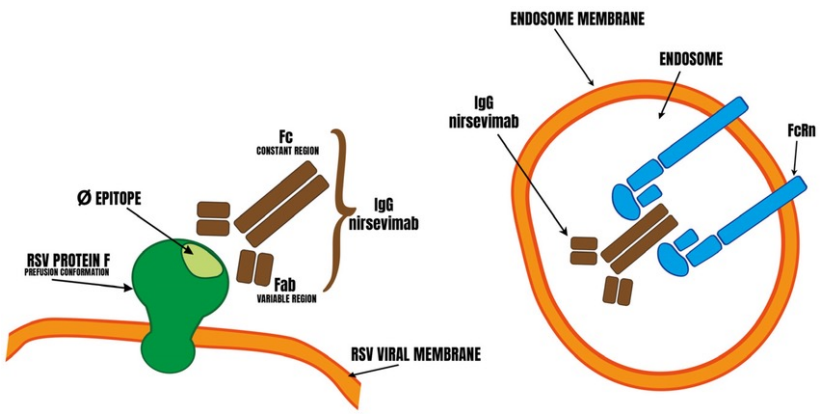
Key Results

55% reduction in RSV hospitalization

Significance

Palivizumab was the first demonstration that a monoclonal antibody could prevent an infectious disease in humans. It established that passive immunization with anti-RSV F-protein IgG1 could meaningfully reduce severe illness, setting the stage for next-generation approaches.

MELODY: Nirsevimab



https://www.mdpi.com/1467-3045/46/9/617

Key Results

Table 2. Medically Attended Lower Respiratory Tract Infections and Hospitalizations Associated with Respiratory Syncytial Virus (RSV) through 150 Days after the Injection.*

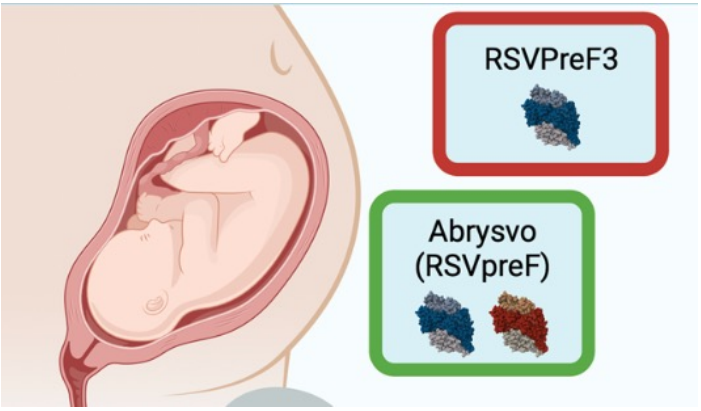
End Point and Analysis	Nirsevimab (N=994)	Placebo (N=496)	Efficacy (95% CI)†	P Value
	no. (%)			
Medically attended RSV-associated lower respiratory tract infection			74.5 (49.6 to 87.1)	<0.001
Poisson regression with robust variance				
Observed events	12 (1.2)	25 (5.0)		
Participants with imputation of data‡	15 (1.5)	6 (1.2)		
Hospitalization for RSV-associated lower respiratory tract infection			62.1 (−8.6 to 86.8)	0.07
Poisson regression with robust variance				
Observed events	6 (0.6)	8 (1.6)		
Participants with imputation of data‡	15 (1.5)	6 (1.2)		

(2)

Advance Over Palivizumab

Nirsevimab's Fc-engineered extended half-life eliminated the need for monthly dosing. A single pre-season injection provided season-long protection and critically, it was studied in healthy term and late-preterm infants, broadening the eligible population beyond the high-risk niche served by palivizumab.

MATISSE: RSVpreF Vaccine (Abrysvo)

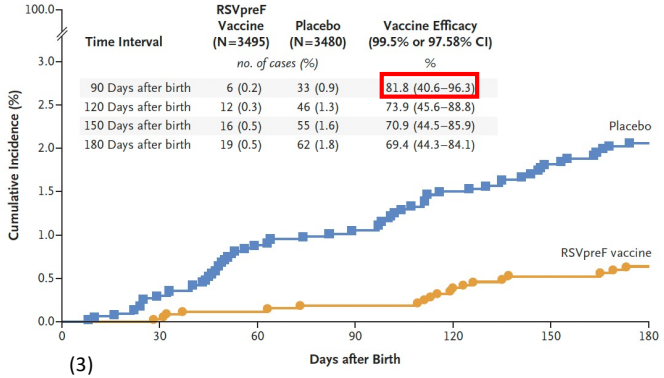


https://www.nature.com/articles/s41541-023-00734-7

Key Results

A Medically Attended Severe RSV-Associated Lower Respiratory Tract Illness

Time Interval	RSVpreF Vaccine (N=3495)	Placebo (N=3480)	Vaccine Efficacy (99.5% or 97.58% CI)
	no. of cases (%)		%
90 Days after birth	6 (0.2)	33 (0.9)	81.8 (40.6–96.3)
120 Days after birth	12 (0.3)	46 (1.3)	73.9 (45.6–88.8)
150 Days after birth	16 (0.5)	55 (1.6)	70.9 (44.5–85.9)
180 Days after birth	19 (0.5)	62 (1.8)	69.4 (44.3–84.1)



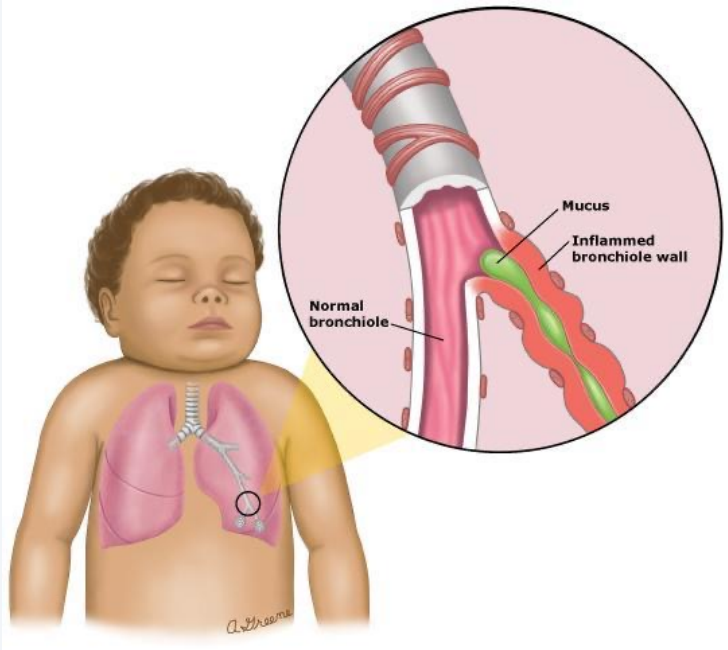
(3)

A New Paradigm

MATISSE demonstrated that a single maternal vaccination could harness natural transplacental IgG transfer to protect infants from birth, the period of greatest vulnerability. This approach leverages the same IgG biology as mAb prophylaxis but through active immunization, potentially offering broader global reach and eliminating the need for infant injection.

Clinical Significance

#1	Leading cause of acute lower respiratory tract illness in infants worldwide
90K+	Pediatric hospitalizations annually in the United States alone
4,500	Estimated annual deaths in U.S. children from RSV complications
2–3 mo	Peak age for severe RSV illness, despite presence of maternal antibodies
~60%	Of RSV hospitalizations in children <1 year occur in infants <3 months



https://www.afterhourskids.com/symptoms/bronchiolitis-and-rsv

Key Takeaways

IgG as the Foundation

All three strategies rely on IgG-mediated protection, whether delivered directly as a monoclonal antibody (palivizumab, nirsevimab) or generated via maternal vaccination and transferred transplacentally (RSVpreF).

25 Years of Progress

From monthly injections for high-risk premature infants (1998) to single-dose, season-long protection for all infants (2022) and maternal immunization protecting from the moment of birth (2023).

Global Equity Imperative

RSV burden is greatest in low- and middle-income countries. Maternal vaccination offers scalable protection without cold-chain-intensive mAb products, addressing health equity gaps.

Future Direction

Working to combine maternal vaccination and single-dose extended half-life mAbs so that severe RSV illness in children can be a preventable condition.

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